

Alert Fatigue in Security Operations Centres: Research Challenges and Opportunities

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A security operations centre (SOC) is a facility where teams of security professionals, supported by advanced technologies and processes, work together to monitor, detect, and respond to cybersecurity incidents. With advances in AI technology, most of the SOC functions are increasingly becoming AI-driven. Among these, real-time alert monitoring and triage is particularly important. Recent studies, by both industry and academia, have highlighted the problem of alert fatigue and burnout in SOC. Several solutions have been proposed in the literature and by the industry to address this problem. In this article, we review the existing literature and industry solutions on alert fatigue mitigation through the lenses of automation, augmentation, and human-AI collaboration. Based on the review, we identify four major causes of alert fatigue in SOC. We also examine the shortcomings of existing solutions and propose several potential research directions leveraging AI. By providing a comprehensive analysis of the state-of-the-art approaches and their limitations, this study contributes to the existing literature in an important field of study. We anticipate that it will inspire new research directions for addressing alert fatigue not just in SOC but across other Command and Control (C2) domains as well.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Security and privacy** → **Network security**; *Intrusion/anomaly detection and malware mitigation*; • **Computing methodologies** → Artificial intelligence; Machine learning; • **Human-centered computing** → *Interactive systems and tools*;

Additional Key Words and Phrases: Security operation centre, SOC, alert triage, alert fatigue, automation, augmentation, collaboration

ACM Reference Format:

Shahroz Tariq, Mohan Baruwat Chhetri, Surya Nepal, and Cecile Paris. 2025. Alert Fatigue in Security Operations Centres: Research Challenges and Opportunities. *ACM Comput. Surv.* 57, 9, Article 224 (April 2025), 38 pages. <https://doi.org/10.1145/3723158>

1 Introduction

In 2022, cybercrime damages cost the world \$8.44 trillion USD, and this figure is expected to surge in the next four years, rising from \$9.22 trillion in 2024 to \$13.82 trillion by 2028 [44]. Cybercrime has increased by 600% since the COVID-19 pandemic, as more business operations shift

This research was financially supported by CSIRO's Collaborative Intelligence Future Science Platform. Figures are designed using icons from Flaticon.com.

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ACM 0360-0300/2025/04-ART224

<https://doi.org/10.1145/3723158>

online [108]. For many businesses, protecting digital assets from cybercriminals remains a major challenge. Many organisations have traditionally used security tools such as firewalls, intrusion detection and prevention systems, and antivirus solutions to safeguard their networks and data from cyberattacks. However, it is no longer sufficient to deploy these defensive technologies [22]. With the recent advancements in big data and AI, cybercriminals are typically one step ahead of the defensive technologies, as they can now generate more sophisticated attack vectors that can evade the majority of defence solutions. For example, attackers can manipulate firewall behaviour or carry out advanced attacks on **intrusion prevention systems (IPSs)**. The main challenge for organisations is thus how to protect their assets from such sophisticated cyberattacks.

Security Operations Centres (SOCs) have emerged as a way for organisations to respond to cyber incidents in a timely, coordinated, and effective manner [77]. The primary function of a SOC is to monitor, detect, investigate, respond to, and protect an organisation from cyber threats. In recent times, all sectors have seen an increase in the need for SOC services. According to the 2017 McAfee Labs Threats Report [11], 94% of big corporations employ a SOC. Similarly, the SANS 2022 SOC Survey found that around 80% of organisations in four key industries (technology, banking, cybersecurity, and government) either have an internal SOC or have contracted external service providers [26]. Interestingly, the majority of the study participants were from small- or medium-sized businesses, demonstrating the high demand for SOC in smaller businesses as well. It is predicted that the global SOC demand will continue to grow from \$6.7 billion USD in 2023 to an estimated \$11.4 billion by 2028 [78].

The effectiveness of SOC hinges on the use of various tools and automation processes to effectively capture, aggregate, enrich, correlate, prioritise, and present security-relevant data to SOC analysts, who perform *alert triage*, the process of determining the validity and severity of security alerts. However, SOC analysts are unable to keep up with the rapidly increasing volume of alerts even with the aid of specialised SOC tools; this is further compounded by the high occurrence of false positives, primarily due to SOC's conservative approach to automated monitoring. A recent Trend Micro survey [83] found that 51% of SOC teams feel overwhelmed by alert volume, with analysts spending over 25% of their time handling false positives. When SOC analysts are under excessive and prolonged stress, it desensitises them and compromises their effectiveness, leading to alert fatigue and burnout. A recent survey by Tines [56] finds that 63% of practitioners experience some level of burnout, with more than 80% saying their workload has increased in the past year. This can lead to real threats being overlooked, posing potentially catastrophic risks to organisations.

The growing adoption of AI-enabled tools in SOC strengthens the case for human-AI collaboration [106, 116], also referred to as hybrid intelligence [2]. The recently proposed A^2C Framework [21] envisions a future where humans and AI work seamlessly to address complex cybersecurity challenges. This framework advocates for flexible autonomy wherein the level of automation can be dynamically adjusted over time. It supports three distinct modes of decision-making: *Automation* addresses the challenge of scale in SOC by efficiently managing routine tasks; *augmentation* enhances human decision-making by providing additional information or context; *human-AI collaboration* (or collaborative exploration) leverages the complementary strengths of humans and AI to improve detection and response to novel security incidents. In this context, the A^2C lens, i.e., *automation*, *augmentation*, and *collaboration*, offers a useful framework for examining the various approaches proposed in the literature to mitigate alert fatigue.

CONTRIBUTION. This article systematises solutions for alert fatigue in SOC, examining them through the A^2C lens.

- (1) **Categorisation of Alert Fatigue Causes:** We categorise the primary causes of alert fatigue in SOC.

Table 1. Comparison of Contribution between Our Work and Relevant Studies, Surveys, and Literature Reviews

References	Year	SOC Challenges	Studies Alert Fatigue	Reviews Alert Fatigue Solutions	Research Method
Dykstra and Paul [35]	2018	✓	✓	✗	Qualitative Study
Onwubiko and Ouazzane [101]	2019	✓	✗ (Limited)	✗	Literature Reviews
Kokulu et al. [66]	2019	✓	✗ (Limited)	✗	Qualitative Study
Vielberth et al. [141]	2020	✓	✗ (Limited)	✗	Literature Reviews
Trend Micro [83]	2021	✓	✓	✗	Industry Survey
Alahmadi et al. [3]	2022	✓	✓	✗	Qualitative Study
IBM [86]	2023	✓	✗ (Limited)	✗	Industry Survey
SANS [26] (Yearly)	2016–2023	✓	✗ (Limited)	✗	Industry Survey
Tines [56] (Yearly)	2022–2023	✓	✗ (Limited)	✗	Industry Survey
Ours	2024	✓	✓	✓	Literature Reviews

- (2) **Systematic Review of Alert Mitigation Solutions:** We systematise the recent literature and industry solutions on alert fatigue mitigation in SOC through the A^2C lens and discuss the shortcomings of existing approaches. We also map the causes of alert fatigue in SOC to the proposed solutions.
- (3) **Future Research Directions:** We identify several potential research directions for mitigating alert fatigue in SOC, discuss associated challenges, and offer actionable recommendations to address them.

ALIGNMENT WITH INDUSTRY TRENDS AND DEMANDS. This work strategically focuses on human–AI collaboration solutions, a concept gaining traction in SOC [21, 106, 143], particularly with the rise of LLMs. [102]. A Microsoft study [36] found that 93% of security analysts are open to the idea of working alongside an AI teammate.

BROADER APPLICABILITY. Alert fatigue is not exclusive to SOC. It is a challenge in various **Command and Control (C2)** domains where operators face high-volume alerts, including defence C2 centres [48], emergency response centres [5], satellite control rooms [136], factory automation systems [31], and energy control rooms [28], to name a few. Our work informs the development of more effective solutions across a broader range of C2 domains.

ARTICLE ORGANISATION. The rest of the article is structured as follows: Section 1 introduces this work and contributions. Section 2 provides an overview of the related work. Section 3 discusses our paper collection methodology. Section 4 provides an overview of SOC and its evolution. Section 5 discusses the process of alert validation, the problem of alert fatigue, and provides a categorisation of four primary alert fatigue causes. Sections 6 and 7 present a comprehensive taxonomy of alert fatigue mitigation approaches from both academic research and industry practices, categorising them under automation, augmentation, and collaboration. Each category is discussed in detail, highlighting the strengths and limitations of the proposed approaches. Section 8 maps the identified literature and industry solutions to the four causes of alert fatigue. Section 9 outlines the key future research directions, and Section 10 concludes the article.

2 Related Studies

In recent years, numerous studies have focused on exploring the challenges in SOC. Table 1 compares these studies, examining whether they: (i) address SOC challenges, (ii) explore alert fatigue in SOC, and (iii) review alert fatigue solutions for SOC in the literature. In this section, we provide a brief overview of these studies.

- **LITERATURE REVIEWS.** Literature reviews, such as the one by Onwubiko and Ouazzane [101], highlight inefficiencies in SOC due to factors such as insufficient monitoring feeds, excessive dashboards, SOC maturity, and complex onboarding. Similarly, Vielberth

et al. [141] employ the **People-Process-Technology (PPT)** framework to categorise SOC-related concepts and highlight several challenges, including monotonous tasks, staffing shortages, and automation deficiencies. While these studies offer a foundational understanding of the broader landscape of SOC challenges, they do not delve into the proposed solutions. Our focus is on this aspect, with a specific emphasis on the challenge of alert fatigue.

- **QUALITATIVE STUDIES.** Several qualitative studies have shed light on the inherent complexities within SOCs. For instance, Dykstra and Paul [35] introduced a cost-effective method to assess fatigue, frustration, and cognitive workload. Kokulu et al. [66] conducted semi-structured interviews that revealed mismatches in understanding and expectations between SOC managers and analysts. These studies underscore the complexity of SOC operations and highlight risks posed by technological shortcomings, such as poor log quality and information overload. Alahmadi et al. [3] conducted qualitative research with 21 SOC analysts, emphasising the laborious nature of alarm validation and its detrimental effects on SOC efficiency. While these qualitative insights are valuable, our research focuses on consolidating proposed solutions to mitigate alert fatigue in SOCs.
- **INDUSTRY SURVEYS.** Industry surveys, such as those by Trend Micro [83], the SANS Institute [26], Tines [56], and IBM [86], offer global insights into SOC operations. The SANS Institute and Tines have annually surveyed SOCs since 2016 and 2022, respectively, to assess capabilities, deployment strategies, technology usage, and challenges. Both have identified alert fatigue and analyst burnout as top challenges for SOCs. Trend Micro and IBM have examined the impact of inadequate tooling on security analysts. Trend Micro found that 51% of SOC teams are overwhelmed with alert volume, whereas IBM reported that teams were only able to resolve 49% of alerts assigned to them in a workday. These surveys provide valuable data on industry trends and challenges faced by SOC practitioners. They do not, however, connect these challenges to solutions in the literature. This is where our work becomes relevant.
- **OUR STUDY.** Our study diverges from the problem-centric approach commonly found in prior studies and adopts a solution-oriented perspective. Specifically, we focus on solutions addressing the challenge of alert fatigue. We categorise existing solutions through the A^2C lens, providing a more comprehensive understanding of the potential benefits and limitations of employing AI for alert triage in SOCs. We anticipate that our study will inform future research and development efforts aimed at mitigating alert fatigue in SOCs, ultimately enhancing the effectiveness and efficiency of alert management. Furthermore, we believe that this work has the potential to extend beyond the SOC, fostering the development of general solutions for mitigating alert fatigue across a wider range of C2 domains.

3 Paper Collection Methodology

We conducted a systematic literature review following the methodology developed by Wolfswinkel et al. [146], comprising five steps as outlined below: (i) *Define*, (ii) *Search*, (iii) *Select*, (iv) *Analyse*, and (v) *Present*. The use of a rigorous and systematic approach ensures the validity of our survey article.

- **DEFINE.** This stage involves clearly defining the scope and objectives of the survey and establishing criteria for selecting relevant literature. Our objective was to analyse a range of academic literature and other high-quality peer-reviewed sources on alert fatigue solutions in SOCs. We used the following criteria to identify relevant papers:

- *Inclusion Criteria.* The paper must (i) be a peer-reviewed journal article, conference/workshop paper, or book chapter¹; (ii) discuss alert fatigue in SOC(s); and (iii) discuss the use of technology to mitigate alert fatigue.²
- *Exclusion Criteria.* Papers related to SOC(s) that did not address the problem of alert fatigue were excluded.
- **SEARCH.** This stage entails a comprehensive search of academic databases, professional forums, and industry publications. We used Google Scholar (*scholar.google.com*) as the primary source for papers due to its extensive coverage of research topics across a variety of fields [79]. We used a combination of keywords such as “alert fatigue,” “alarm fatigue,” “SOC(s),” and “security operation(s) centre(s) or center(s)” to ensure that we captured relevant papers. We set the search period to 2017–2024 and used the snowballing approach to identify additional papers.
- **SELECT.** The *Select* stage involves applying the *inclusion* and *exclusion* criteria to narrow down the list of articles. The papers discovered during the search phase were screened based on their titles and abstracts to select the ones that met the pre-established inclusion criteria. We included papers that discussed solutions to mitigate alert fatigue in SOC(s). We excluded papers that were not written in English, did not provide enough details on the proposed solution, or were not published in reputable journals or conferences.
- **ANALYSE.** This stage involves critically evaluating the selected literature. We categorised the papers by solution type and examined their strengths and limitations. For instance, we looked at papers proposing automated solutions (e.g., machine learning), human augmentation tools (e.g., dashboards or decision support systems), and human–AI collaboration. We evaluated the quality of the papers using a set of predefined criteria, such as methodological rigour, the reliability of the data presented, and the relevance of the findings to the context of this work.
- **PRESENT.** Finally, the *Present* stage involves organising and presenting the findings of the survey clearly and concisely. Papers are organised under three broad categories: automation, augmentation, and collaboration. We have included a summary of each paper, covering its key findings, methodology, and limitations.

4 Security Operations Centre Overview

A SOC is a team of people, processes, and technologies that monitor an organisation’s systems and networks to detect and respond to cybersecurity threats [91]. Key SOC functions include:

REAL-TIME ALERT MONITORING AND TRIAGE. SOC(s) continuously monitor security alerts from tools such as **intrusion detection systems (IDS)**, firewalls, endpoint protection platforms, and **security information and event management (SIEM)** systems. Analysts assess alerts to distinguish false positives from legitimate threats, prioritise them based on severity, and initiate responses to mitigate security risks [90].

INCIDENT ANALYSIS AND RESPONSE. Analysts investigate confirmed or suspected security incidents to determine their scope, impact, and root cause. Using techniques such as log analysis, malware analysis, memory forensics, and network packet inspection, they gather forensic evidence and implement response strategies, including containment, eradication, and recovery, to minimise damage [141].

¹We first attempted to find relevant articles in top venues. If we could not find enough articles there, then we looked for articles in lower-tier venues.

²To provide a comprehensive context, we included relevant studies on SOC(s) and alert fatigue even if they did not fully meet our inclusion criteria.

INCIDENT COORDINATION. SOC managers manage communication among internal teams (e.g., IT, legal) and external entities (e.g., law enforcement, third-party vendors) during incident response. This includes documenting incidents, maintaining evidence integrity, and ensuring regulatory compliance. Effective coordination streamlines response efforts and enhances situational awareness across stakeholders [65, 75].

In addition to core functions, modern SOC managers perform specialised tasks to strengthen cyber defences. These include in-depth forensics, where analysts reconstruct security incidents, identify attackers, and gather legal evidence; threat hunting, which involves proactively detecting hidden threats using advanced analytics and threat intelligence; attack simulations, such as penetration testing and red team exercises, to evaluate security controls and enhance incident response readiness; and deception technologies, where techniques such as honeypots and decoy assets lure attackers, providing insights into their tactics. These proactive measures enhance threat detection, incident response, and overall security posture against evolving cyber threats [89], complementing traditional SOC functions.

4.1 SOC Structure

SOC structures have been classified in various ways. Kowtha et al. [68] identified seven functional dimensions: scope, activities, process management, facilities, organisational dynamics, external interactions, and environment. Another widely used framework is the **People, Processes, and Technology (PPT)** model [141], which emphasises the need for skilled IT specialists (people) to operate security measures (technology) using standardised workflows and best practices (processes). Recently, Vielberth et al. [141] extended this model by adding Governance and Compliance, which were previously considered part of processes. Other classifications focus on the people aspect. Hámornik and Krasznay [52] examined SOC team structures to understand decision-making under high-stress conditions. Analysts are typically organised into three tiers based on expertise, with tasks escalating from lower to higher tiers according to severity. Teams rely on two types of knowledge: declarative knowledge (e.g., “what to do” in a known cybersecurity incident, shared among all members) and procedural knowledge (e.g., “how to do” specific tasks, which is experience-based, non-verbalised, and not universally possessed within the team).

4.2 Evolution of SOC

SOCs must continually evolve to keep pace with the dynamic cyber threat landscape. Figure 1 illustrates the evolution of SOC managers, highlighting key features and their alignment with Automation, Augmentation, and Collaboration.

FIRST-GENERATION SOC. The **Network Operations Centre (NOC)** was the precursor to the modern SOC. Its primary responsibilities included performance monitoring, malicious code analysis, and network device management [51]. Until the mid-1990s, NOCs were predominantly utilised by telecommunications, government, and military entities.

SECOND-GENERATION SOC. They relied on reactive measures such as firewalls, IDS, and antivirus tools [38]. They were adopted by government, military, large corporations, and financial institutions from the mid-1990s to early 2000s.

THIRD-GENERATION SOC. Third-generation SOC managers added features such as vulnerability management, packet filtering, spam filtering, and intrusion prevention [13]. They were reactive systems operating from the early 2000s to 2006.

FOURTH-GENERATION SOC. Popular during 2007–2012 and adopted by medium-sized businesses as well [12], these SOC managers incorporated an early version of SIEM and concentrated on locating advanced persistent threats through the use of threat intelligence, anomaly detection, and malware analysis [112].

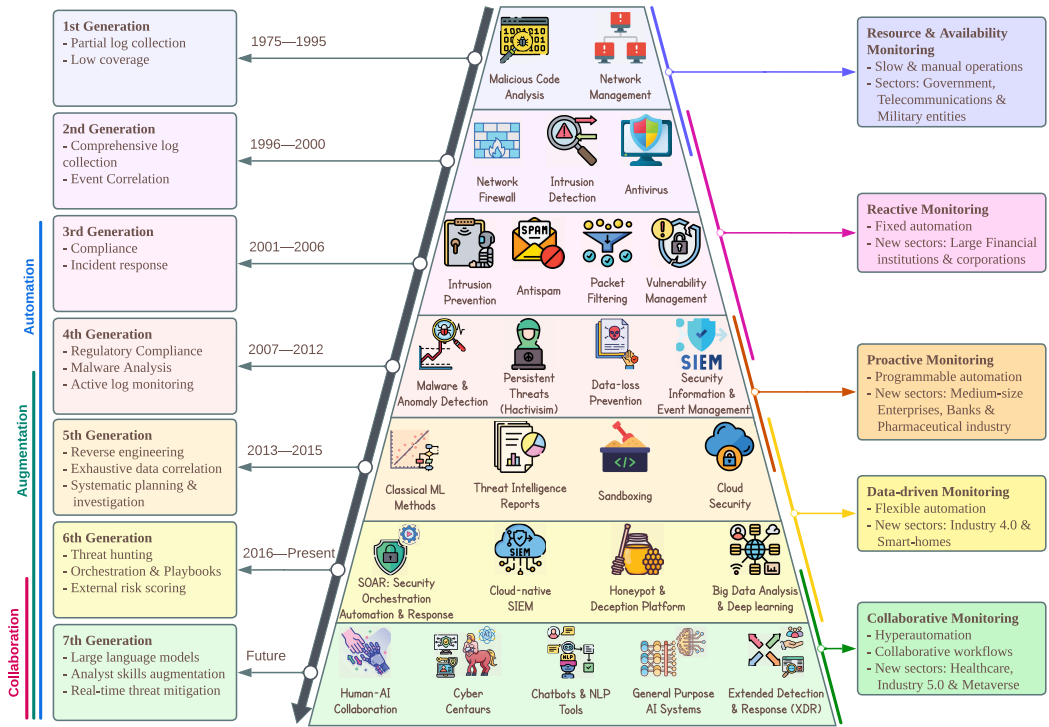


Fig. 1. Evolution of SOCs: From manual to collaborative monitoring and the rise of automation, augmentation, and collaboration.

FIFTH-GENERATION SOC. Also known as Hybrid SOCs, the fifth generation of SOC incorporated cloud-based security, AI/ML models, and **user and entity behaviour analytics (UEBA)** on top of existing SIEM technology to reduce false positives [90]. They also made use of **open-source intelligence (OSINT)** and **threat intelligence platforms (TIPs)**. During 2013–2015, SOCs started being used in the industrial sector as well.

SIXTH-GENERATION SOC. The current generation of SOC employs big data analysis, intelligence-driven methodologies, and automation to detect threats. This paves the way for the implementation of **Security Orchestration, Automation, and Response (SOAR)** and playbooks [141].

SEVENTH-GENERATION SOC. Looking at the trends from previous generations of SOCs and the rise of AI/ML, particularly generative AI, one can imagine that the next generation of SOCs will be more centred on collaboration, with a more interactive human and AI ecosystem. **Extended detection and response (XDR)** is another emerging direction [63].

5 Alert Triage

Effective SOCs rely on three key aspects: (i) *SOC monitoring platform*, which handles log collection, aggregation, analysis, and incident management; (ii) *cyber onboarding*, the process of integrating new and existing services into the monitoring platform; and (iii) *eyes-on-glass*, where analysts continuously monitor alerts using structured processes and technology [100]. As part of eyes-on-glass monitoring, SOC analysts perform alert triage, assessing the relevance, severity, and accuracy of security alerts. Alert fatigue occurs when excessive alerts overwhelm analysts, leading to desensitisation, ignored alerts, and potentially missed threats or security breaches. This section provides an overview of alert triage, alert fatigue, and its underlying causes.

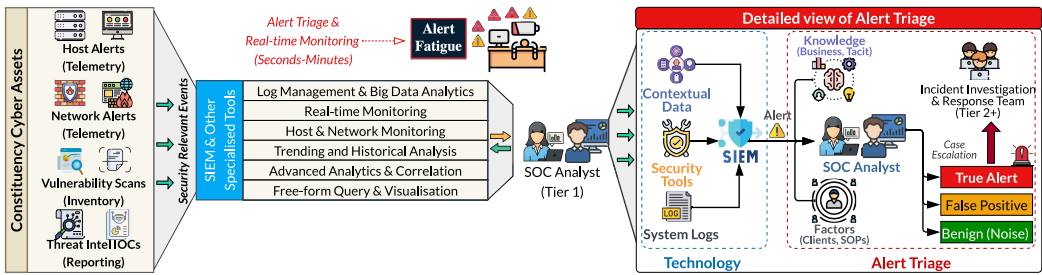


Fig. 2. A detailed view of the alert triage workflow in security operation centres.

5.1 Alert Triage Process

Figure 2 presents a simplified overview of the alert triage process, while a broader SOC workflow can be found in References [65, 89]. Security telemetry from an organisation’s cyber assets is continuously collected and pushed to a central tool, such as a **security information and event management (SIEM)**, along with contextual data. The SIEM aggregates, enriches, and correlates security events—observable occurrences within the system or network—to generate alerts, which notify SOC analysts of potential threats. Alerts may also originate from security tools like network/host intrusion detection systems. A key SOC analyst task, *alert triage*, involves sorting, categorising, and prioritising alerts using advanced analytics, correlation techniques, queries, and visualisations. Analysts leverage both system knowledge and tacit experience to determine whether an alert is legitimate or a false positive. Verified threats are escalated to the incident response team.

SOC analysts’ performance is impacted by the *volume*, *velocity*, *veracity*, and *variety* of alerts. The high rate (*velocity*) and sheer number (*volume*) of alerts increase the cognitive burden, while SIEMs—often tuned to minimise false negatives in intrusion detection-related components—produce excessive false positives (*veracity*). Additionally, the evolving nature of cyber threats (*variety*) challenges ML and heuristics-based solutions, further contributing to false positives and analyst fatigue. SIEM correlation rules that aggregate multiple indicators improve alert reliability, but their effectiveness is limited by the capabilities of the conventional security tools that generate the alerts. Notably, not all alerts are false positives; some benign triggers match existing signatures but may be disregarded for “business-justified” reasons [3]. However, alert triage remains a repetitive and exhausting task, leading to analyst burnout.

5.2 Alert Fatigue

The Trend Micro survey [83] found that 54% of SOC teams feel overwhelmed by alerts, 55% lack confidence in prioritising or responding to them, and security experts spend 27% of their time handling false positives. The 2022 SANS SOC Survey [26] reported similar concerns, with analysts stating they face “too many alerts” and insufficient correlation between them. The 2023 Times survey [56] identified excessive alerts as a leading cause of alert fatigue and burnout, while IBM’s SOC survey [86] listed false positives as one of the top four challenges, with 76% of SOC teams attributing delays in threat response to low-priority alerts. These industry reports highlight that, despite mitigation efforts, enterprises continue to struggle with alert fatigue, information overload, false positives, and burnout [42].

Research further emphasises the human impact of alert fatigue. Sundarmurthy et al. [134] identified exhaustion, cynicism, and inefficiency as key factors contributing to burnout and high turnover among SOC analysts. Dykstra and Paul [35] found that analysts experience significantly

higher stress, fatigue, and frustration during cybersecurity operations. Husák et al. [58] categorised challenges in achieving cyber situational awareness and concluded that the increasing volume, velocity, and variety of cybersecurity data exacerbate exhaustion. Alert fatigue extends beyond SOC, affecting hospital emergency rooms [5], construction sites [7], and even COVID-19 protection compliance [145], where excessive alerts led to desensitisation and non-adherence.

5.3 Causes of Alert Fatigue

Through a meticulous evaluation of the existing literature [3, 65, 66, 89, 141], including industry survey papers [11, 26, 83], we have identified four overarching groups of factors causing alert fatigue in SOC. It should be noted that these categories are the most prevalent causes of alert fatigue, although other minor factors also contribute to the problem. We provide a concise overview of each of these categories. Understanding and addressing these factors is crucial for improving the efficiency and effectiveness of SOC.

5.3.1 Staff and Skills Shortage. One of the primary obstacles that SOC face is the significant demand for skilled personnel and the shortage of proficient analysts [26]. This leads to long working hours, overload, and, ultimately, alert fatigue. The issue is exacerbated by the continuous growth in the volume and complexity of security alerts generated by various security tools and systems. These alerts demand immediate attention, thereby placing additional cognitive strain on SOC analysts.

5.3.2 High False-alarm Rate. One of the biggest problems with SOC is the high rate of false positives [83]. A pressing challenge for SOC is to develop effective strategies and tools to reduce alert volume and improve the accuracy of correlation processes. The design and implementation of such tools are complex endeavours that require careful attention to diverse factors affecting SOC functionality, including alert type, volume, and severity.

5.3.3 Disconnected and Overloaded Dashboards. SOC struggle with fragmented and overloaded dashboards, hindering enterprise-wide visibility and alert correlation [26]. The lack of visibility stems from analysts' restricted access, difficulties in consolidating data from multiple sources, and the absence of tools for organising and presenting information. This complicates the correlation of alerts from different tools, hindering the identification of multi-stage and complex attacks. Moreover, the lack of integration between tools reduces efficiency in managing data, further impacting SOC analysts' effectiveness.

5.3.4 Inefficient Standard Operating Procedures (SOPs). The challenge of alert fatigue is further aggravated by inefficient SOPs, which comprise formal, written instructions for incident response, including both operational and technical components. As reported in the SANS SOC Survey [26], the lack of automation and orchestration, as well as the absence of well-defined processes and playbooks, prevents SOC from maximising their capabilities. The manual nature of SOPs necessitates interventions that are prone to errors, delays, and inconsistencies. The lack of orchestration results in poor coordination and communication between teams and tools involved in incident response. Similarly, the absence of structured playbooks diminishes SOC's ability to consistently and effectively respond to security incidents. All of these inefficiencies further intensify alert fatigue.

6 Alert Fatigue Mitigation Solutions in Literature

The literature presents various approaches to mitigating alert fatigue in SOC, typically addressing only a subset of the underlying causes outlined in Section 5.3. Each solution is analysed based on eight factors: (i) methodology, (ii) baselines, (iii) dataset, (iv) metrics, (v) performance, (vi) strengths

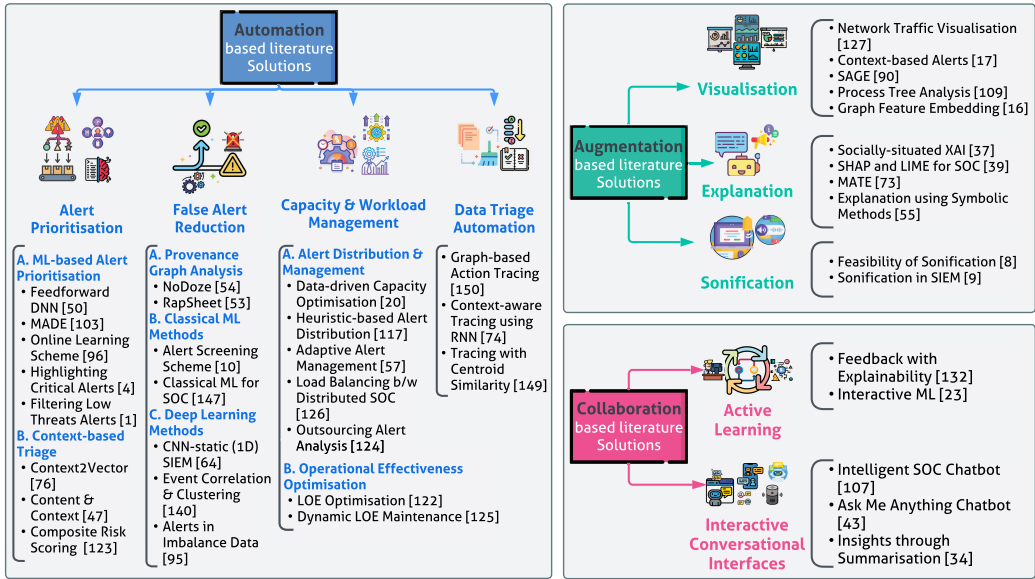


Fig. 3. Systematic categorisation of academic solutions on alert fatigue through the A²C lens.

(Pros), (vii) weaknesses (Cons), and (viii) potential improvements (future work). This information is then condensed into comparison tables (Tables 2–7) using four key columns: (i) method name and main technique, (ii) dataset used for training and evaluation, (iii) performance and corresponding metric, and (iv) remarks on pros, cons, and future work.

6.1 Automation-based Alert Fatigue Mitigation Solutions

Automation is the most common approach for mitigating alert fatigue in SOCs. In our study, we have categorised automation-based methods into four primary categories (see Figure 3): (i) Alert Prioritisation, (ii) False Alert Reduction, (iii) Capacity and Workload Management, and (iv) Data Triage Automation.

6.1.1 Alert Prioritisation. Automation provides a framework for prioritising alerts based on severity and relevance. Several alert prioritisation solutions in the literature rely on automation (see Table 2). We categorise these solutions into two groups: (A) ML-based Alert Management and (B) Context-based Alert Ranking.

A. Machine Learning-based Alert Management. Gupta et al. [50] propose an automated event prioritisation approach for SOCs using a feedforward deep neural network. They identify features through graphical analysis and feature merging, employing filters and embedded methods to mitigate overfitting during training. However, this method introduces processing overhead and requires optimisation. Future work should focus on improving feature engineering efficiency. One potential direction is incorporating additional graph metrics, such as diameter and eccentricity, to enhance model robustness. Oprea et al. [103] introduce **Malicious Activity Detection in Enterprises (MADE)**, a supervised ML method using the **Random Forest (RF)** algorithm to analyse security log data and detect malicious activities in enterprise networks. MADE prioritises detected activities based on their risk levels, leveraging 89 features identified by SOC domain experts. It effectively identifies high-risk domains contacted by enterprise hosts with high precision and a low false-positive rate. However, it struggles with extremely stealthy attacks,

Table 2. A Summary of Automation-based Alert Fatigue Mitigation Solutions Focusing on Alert Prioritisation

Method (Name; Technique)	Dataset	Performance (Metric)	Remarks (Pros; Cons; Future directions)
N: Feedforward DNN [50] T: DNN	Event Logs	92% (F1)	P: Alert fatigue reduction C: DNN complexity F: Graph-based feature engineering
N: MADE [103] T: Classical ML (LR, SVM, RF, DT)	Web proxy logs	97% (Precision)	P: Prioritises connections C: Miss advanced persistent threats F: HTTPS malicious detection
N: Online Learning Scheme [96] T: Focal Loss with logistic regression	Class-B network traffic	98% (Accuracy)	P: Low false-positive rate C: Relies on labelled data F: Semi-supervised learning
N: Highlighting Critical Alerts [4] T: Isolation Forest, Stacked Auto Encoder	Class-B network traffic	96% (Recall)	P: Low false-positive rate C: Real-time processing complexity F: Method generalisation
N: Filtering Low-threat Alerts [1] T: Extended Isolation Forest	IDS 2018 Intrusion CSVs	96% (Recall)	P: Monitors 17.85% of alerts C: Discards few anomalies F: Reduce anomalous alerts
N: Context2Vector [76] T: Context representation learning	Enterprise scale dataset	2.25× (Recall)	P: Accelerates triage C: Scalability issues F: Improve interpretability/stability
N: Content & Context [47] T: Predicting alert- and incident-level actionability	Security Alerts	77% (AUC)	P: Suppresses 54% of FPs C: Generalisability issues F: Automatic featurisation
N: Composite Risk Scoring [123] T: Computing a composite risk-score	Simulated environment	Not specified	P: Handles mission-critical alerts C: Two-step process complexity F: Test with real-world datasets

such as **advanced persistent threats (APTs)** and is also vulnerable to adversarial attacks, where attackers manipulate high-importance features to alter classification results. Future research should focus on capturing stealthy, adaptive, and adversarial attacks.

Ndichu et al. [96] highlight the need for frequent retraining or model updates due to the dynamic nature of threat alerts. Batch ML-based analysers face several issues: time-consuming updates, large storage requirements, and poor scalability. To address these, the authors propose an online learning scheme for critical threat alert detection. This scheme treats threat alert data as a stream, allowing for prompt responses to new alerts. They employ a focal loss function to handle class imbalance in threat alert data and report that logistic regression with focal loss surpasses other baseline methods in performance. However, the approach requires labelled data, which is difficult to obtain. Additionally, critical alerts are rare and often evaded. Incorporating an automatic labelling scheme could improve performance.

Aminanto et al. [4] propose using **Isolation Forest (IF)**, a tree-based anomaly-detection algorithm that uses binary trees to detect anomalies by identifying data points with short paths to identify anomalous alerts, combined with a stacked autoencoder to learn underlying patterns in alert data. This approach identifies critical alerts with minimal false negatives, reducing the number of less critical alerts shown to security operators. However, the method focuses on a single security module, while SOC's consist of multiple interacting components. Detection accuracy could be improved by incorporating additional features. Further research is needed to assess the technique's compatibility with various security appliances and log formats. Ahmed et al. [1] use an **Extended Isolation Forest (EIF)**, an extension of the IF algorithm that improves isolation by considering feature correlations, to filter out low-threat alerts, aiming to prioritise critical alerts and reduce alert volume. However, the need for retraining with each new alert makes the model impractical for real-time environments, particularly in high-traffic scenarios.

B. Context-based Alert Ranking. Liu et al. [76] argue that context offers a more precise characterisation of network alerts than traditional rules- and intelligence-based methods, noting

that same-category events can have different objectives in different contexts. They propose Context2Vector, a context representation learning technique that gathers contexts from various behavioural aspects and uses human-in-the-loop topic annotation for event risk ranking. This approach improves alert prioritisation and reduces false alerts, enhancing triage efficiency. However, its reliance on human inspections limits adaptability in dynamic network environments. Developing an automated parameter selection mechanism could mitigate this issue and reduce reliance on human input. Additionally, the proposed contextual characteristics may not cover all possible contexts. Investigating more general and fine-grained context candidates, potentially with LLMs, could provide a deeper understanding of event patterns and their contextual relationships.

Gelman et al. [47] propose an ML framework for predicting alert-level and incident-level actionability through a feature extraction module and an ensemble of ML models. These models analyse feature content and context to generate an alert-level score used by the triage module for incident-level risk scoring and alert prioritisation. The authors consider four algorithms: logistic regression, RF, XGBoost, and a feed-forward neural network. While the proposed system significantly reduces response time for actionable incidents, suppresses false positives with high detection rates, and reduces the number of alerts analysts need to investigate, the framework's performance could be further enhanced by considering generalisable **deep learning (DL)** techniques. Shah et al. [123] propose a two-step approach to improve SOC triaging. First, they compute a composite risk score for each alert using a quantitative value function hierarchy. Second, they use an optimisation model to select alerts for analysis, aiming to maximise SOC performance metrics within given demand and resource constraints. The study assumes all alerts can be assigned a risk score, which may not always be accurate. Additionally, it focuses on optimising alert investigation post-triage, leaving room to optimise the triaging process itself. Future research should explore ML and other advanced techniques to improve risk-scoring accuracy and evaluate their practical impact on SOC operations.

6.1.2 False Alert Reduction. A significant proportion of alerts generated by SOC tools, as high as 40% [150], are false positives. The implementation of automation can play a critical role in reducing false alerts by eliminating extraneous noise and identifying false positives. Various techniques have been proposed to address this issue (see Table 3), which we categorise into three groups: (A) Provenance Graph Analysis, (B) Classical ML Methods, and (C) DL Methods.

A. Provenance Graph Analysis is a technique for tracing the lineage of data by constructing graphs that represent its transformations, sources, and dependencies. Hassan et al. [54] propose NoDoze, which automates alert triage using contextual and historical information to develop causal dependency graphs. These graphs receive anomaly scores based on event frequency, where an aggregate score ranks true alerts higher than false alerts, effectively reducing false positives. In subsequent work, Hassan et al. [53] introduce **tactical provenance graphs (TPGs)** to capture causal dependencies between threat alerts, using a threat-scoring algorithm to rank truly malicious events higher than false alerts and reduce system log storage overhead. NoDoze and TPGs rely heavily on historical data, making them less effective against adaptive attacks or new threats. Future work could involve implementing an anomaly propagation algorithm during dependency graph generation, integrating distributed graph processing frameworks, and leveraging real-time data and ML algorithms to enhance alert ranking accuracy. Additionally, **natural language processing (NLP)** techniques could analyse threat intelligence feeds [99] and social media to identify emerging threats, and incorporate more contextual information about organisational assets and business criticality to further reduce false positives.

B. Classical ML methods encompass conventional approaches such as linear regression, logistic regression, **support vector machines (SVMs)**, RFs, and **k-nearest neighbours (k-NN)**. Ban

Table 3. A Summary of Automation-based Alert Fatigue Mitigation Solutions Focusing on False Alert Reduction

Method (Name; Technique)	Dataset	Performance (Metric)	Remarks (Pros; Cons; Future directions)
N: NoDoze [54] T: Provenance graph analysis	Systems Events and Alerts	84% (False-alarm Reduction)	P: Reduces false positives C: Dependency on historical data F: Event frequency database
N: RapSheet [53] T: Causal dependencies between endpoint detection and response (EDR) generated alerts	Symantec EDR	Better TPG ranking	P: Accelerates investigation C: TPG complexity F: Improve threat scoring method
N: Alert Screening Scheme [10] T: AI-assisted alert screening	Network IDS	99% (F1)	P: Low false positives C: Relies on manual labels F: Semi-supervised methods
N: Classical ML for SOC [147] T: Framework for ML-based analytics	Not Specified	Not Specified	P: Framework to reduce fatigue C: No evaluations F: Evaluate on real-world datasets
N: CNN-static (1D) SIEM [64] T: Convolutional neural network and LSTMs	Security Threat Events	95% (Recall)	P: Cost-effective C: Overhead in large event sets F: Optimise accuracy vs. efficiency
N: Event Correlation & Clustering [140] T: Supervised learning	Lastline and HDFS datasets	91% (Workload Reduction)	P: Automates event analysis C: Relies on Context Builder F: Improve Context Builder
N: Alerts in Imbalance Data [95] T: Cleaning false alerts, oversampling true alerts	Class-B network traffic	99% (F1)	P: Reduces manual auditing C: Decreases precision F: Recall and precision tradeoff

et al. [10] present an alert screening scheme with three primary modules: (i) an alert generation module that collects data from various network-based IDSs; (ii) a feature processing module that merges logs into a simplified format and encodes them as numerical vectors; and (iii) an ML module that uses supervised learning on labelled vectors to construct a prediction model. However, the model’s effectiveness on additional datasets remains unknown, as it has only been validated on a single dataset. Yeshwanth et al. [147] compared several ML models, including one-class SVM, K-means clustering, RF, and decision trees, to identify the most effective approach. They highlight two key points: (1) Unsupervised ML algorithms can model typical network behaviour and detect anomalies without labelled datasets, making them effective for identifying new threats. (2) However, they tend to generate more false positives than supervised methods, impacting system efficacy.

Classical ML algorithms may be insufficient for handling increasingly complex security threats. Advanced algorithms, such as DL, could enhance the system’s ability to handle complex threats. Additionally, continuous learning techniques [29] that can update the model in real time could improve adaptability to changing threat landscapes.

C. Deep Learning Methods encompass advanced techniques utilising artificial neural networks to learn from data and make informed predictions or decisions. Such methods include but are not limited to **Convolutional Neural Networks (CNNs)**, Deep Autoencoder, **Gated Recurrent Units (GRU)**, **Long Short-Term Memory (LSTM)**, and **Recurrent Neural Networks (RNNs)**. Kim and Kwon [64] highlight the limitations of signature-based threat classification models in SIEM systems, which use predefined rules to minimise false negatives but result in many false positives. To address this, they propose DL-based solutions to prevent analyst alert fatigue and tackle zero-day attacks. They developed a 1D CNN-static model and compared it to other models, including CNNs, LSTMs, and GRUs, using a dataset of 2.6 million threat events from a South Korean vendor. Although the authors suggest rebuilding the model for zero-day attacks, this may not be practical in all scenarios. Future research should explore continuous lifelong learning [29] or **active learning (AL)** [110] systems, which may be more effective for addressing new or zero-day attacks.

Table 4. A Summary of Automation-based Alert Fatigue Mitigation Solutions Focusing on Capacity and Workload Management

Method (Name; Technique)	Dataset	Performance (Metric)	Remarks (Pros; Cons; Future directions)
N: Data-driven Capacity Optimisation [20] T: Probabilistic Cyber-alert Management	Security Alerts data	0 (Uncovered True Alerts)	P: Reduces uncovered true alerts C: Optimisation complexity F: Enhance scheduling robustness
N: Heuristic-based Alert Distribution [117] T: Game-theoretic approach	Simulated Stackelberg game	100× Faster Heuristic	P: Improves alert assignments C: NP-hardness F: Develop scalability heuristics
N: Adaptive Alert Management [57] T: Reinforcement learning	Simulated Real-time Monitoring	20% (IDoS Reduction)	P: Reduces IDoS risk C: Lacks personalisation F: Adapt to different human operators
N: Load Balancing in Distributed SOC [126] T: Reinforcement learning	Simulated SOC Data	Outperforms other strategies	P: Dynamically reallocates alerts C: Implementation complexity F: Adapt to larger SOC locations
N: Outsourcing Alert Analysis [124] T: Two-step sequential MIP optimisation	Simulated SOC Data	Decrease in Computational Time	P: Higher throughput C: Economical service vs. computational time F: Extend MIP model to outsourcing problems
N: LOE Optimisation [122] T: Reinforcement learning model	Simulated SOC Data	RL beats greedy method	P: Dynamically optimises the LOE C: Relies on simulated data F: Implement the RL model with real-world data
N: Dynamic LOE Maintenance [125] T: Value-based strategy and RL-based strategy	Simulated SOC Data	RL strategy is best performer	P: Maintains optimal LOE C: Tuning issues F: Get meta-principles for alert backlog formation

Ede et al. [140] propose DEEPCASE, a semi-supervised DL solution for automatically correlating security events. This method constructs and clusters attention vectors for each input sequence, allowing security operators to label clusters instead of individual incidents, thus reducing the number of events requiring further investigation. Despite its effectiveness, DEEPCASE has limitations, such as susceptibility to evasion strategies and adaptive attacks, and it only addresses events within the same machine context. Further research is needed to overcome these limitations and enhance the efficacy and robustness of DEEPCASE. Ndichu et al. [95] assert that most alert data consists of false alerts, with only a small proportion being true alerts. A simplistic engine labelling all alerts as the majority class can appear highly accurate but fails to account for class imbalance, leading to poor generalisation. To address this, they employ a neighbourhood cleaning rule to eliminate ambiguous, noisy, and redundant false alerts, and they use the SVM synthetic minority oversampling technique to generate synthetic true alerts. This approach reduces manual auditing and increases recall performance, though at the expense of precision. Future work should explore methods to improve both recall and precision, such as cost-sensitive, one-class, probability tuning, and hybrid algorithms.

6.1.3 Capacity and Workload Management. Human analysts in SOC's are often overwhelmed by the sheer volume of alerts they need to review. A key issue that contributes to alert fatigue is the gross mismatch between the workload demands and the available capacity. Several approaches have been proposed to address this problem (see Table 4). We divide them into two categories: (A) Alert Distribution and Management and (B) Operational Effectiveness Optimisation.

A. Alert Distribution and Management. One of the key challenges faced by SOC's is the effective distribution of alerts among security analysts. In recent years, several proposed methods have emerged in the literature to address this challenge. Chen et al. [20] propose **Probabilistic Cyber-alert Management (PCAM)**, a data-driven bi-level optimisation technique for SOC capacity management. PCAM uses past shift data to schedule analysts, aiming to reduce unhandled alerts and estimate the ideal mix of junior, senior, and principal analysts for day and night shifts within a budget. Similarly, Schlenker et al. [117] propose a **cyber allocation game (CAG)** to optimally distribute incoming alerts among analysts facing a strategic adversary. They recommend a heuristic approach to solve this NP-hard problem, enabling large-scale application with minimal

impact on solution quality. Huang and Zhu [57] address the issue of attackers generating numerous fake attacks to overwhelm analysts. They propose the **Resilient and Adaptive Data-driven alert and Attention Management Strategy (RADAMS)**, which uses **reinforcement learning (RL)** to de-emphasise alerts based on category labels and incorporates human factors such as expertise, stress, and efficiency.

While these methods effectively manage the alert allocation and reduce unhandled alerts, they have limitations. PCAM and CAG do not consider analyst fatigue, which can impact decision-making. RADAMS does not account for spatiotemporal attacks with historical dependencies, which could affect analyst attention. Future research should address these limitations by exploring methods to account for analyst fatigue and consider spatiotemporal attacks.

In large organisations, alert management is typically handled by multiple SOC's in different regions. To address alert fatigue in such setups, Shah et al. [126] propose a distributed RL approach to dynamically re-allocate alerts, ensuring a uniform workload distribution. They developed a centralised RL agent using stochastic dynamic programming to decide the timing, quantity, and location of alert relocation. For centralised SOC's, Shah et al. [124] suggest outsourcing alert analysis to on-demand analysts using a two-step sequential mixed-integer programming optimisation approach to reduce workload during peak periods. However, these approaches have limitations. The distributed RL approach assumes all SOC's can handle alerts from any location, which may not be true due to varying expertise levels. Future work should consider these differences and adapt relocation decisions accordingly. Additionally, the centralised RL agent may not scale well in large environments, suggesting a need for decentralised approaches.

B. Operational Effectiveness Optimisation. Several factors can disrupt a SOC's **level of operational effectiveness (LOE)**, such as increased alert rates, analyst absence, and new alert patterns. SOC managers often make ad hoc decisions to temporarily extend alert analysis time, which may contribute to alert fatigue. To address this, Shah et al. [122] propose an RL-based approach to optimise LOE through better load balancing and resource optimisation. In a follow-up study, Shah et al. [125] explore discarding alerts to restore LOE and relieve analyst overload. They suggest two strategies: a value-based approach using a static optimisation model and an RL-based strategy using a dynamic optimisation model. These solutions aim to (i) minimise alert discards, (ii) maximise analyst utilisation, (iii) determine the optimal time to discard alerts, and (iv) maintain LOE close to the baseline goal. The RL method provides the shortest **average time to analyse alerts (AvgTTA)** while dismissing the fewest alerts under adverse conditions. However, future work should explore predictive baseline models to estimate increased workload and allocate resources accordingly to reduce analyst fatigue. Additionally, maximising analyst utilisation, as proposed by Shah et al. [125], may increase cognitive workload and alert fatigue, necessitating further investigation of human factors.

6.1.4 Data Triage Automation. Data triage is time-consuming. Automating some or all steps is a popular strategy to mitigate alert fatigue [115] (see Table 5 for a summary of data triage automation solutions). Automation can streamline the process by scrutinising incoming data and categorising it based on predefined criteria, such as threat type, risk level, or impacted systems. This reduces analysts' workload and allows them to focus on critical alerts for quicker response. For instance, automation tools can scan network traffic logs, recognise suspicious patterns, and alert analysts to potential threats, expediting incident response and minimising workload. In that context, Zhong et al. [150] propose a computer-assisted graph-based tracing approach to document SOC analysts' actions during the data triage tasks. As analysts perform an operation, rules are derived from the analyst's trace to develop a finite state machine for automating data triage for future events. Further investigation is required to improve the usability and intuitiveness of automation tools and

Table 5. A Summary of Automation-based Alert Fatigue Mitigation Solutions Focusing on Data Triage Automation

Method (Name; Technique)	Dataset	Performance (Metric)	Remarks (Pros; Cons; Future directions)
N: Graph-based Action Tracing [150] T: Graph-based trace mining	Network Monitoring Data	1% (FPR), 22% (FNR)	P: Reduces analysts' workloads C: Lack of customisation F: Improve the automated rule-generation system
N: Context-aware Tracing using RNN [74] T: Deep learning using RNNs	Simulated Test Cases	79% (F1-score)	P: Assists junior analysts C: Manual ground truth labelling F: Explore graph- and ontology-based data triage operation retrieval
N: Tracing with Centroid Similarity [149] T: Context modelling	Data Triage Operations	70% (F1-score)	P: On-the-job suggestions to novice analysts C: Relies on historical traces F: Expand trace collection; refine context modelling

to enable their customisation according to the particular requirements of various organisations. Furthermore, RL has demonstrated its ability to adapt to evolving threat environments [144] and could thus facilitate the development of an intelligent agent that autonomously performs data triage by utilising continuous feedback from SOC analysts.

Lin [74] identify that the complexity and intensity of data triage tasks can challenge novice analysts. To address this, they propose an on-the-job suggestion system that presents relevant data triage operations conducted by senior analysts. They developed a tracing method to track an analyst's operations. The proposed system includes (1) context modelling of the data triage process, (2) an RNN to compare matching contexts, and (3) presenting matched traces as suggestions to novice analysts. The system's performance, evaluated through automated testing and human evaluation, effectively identifies relevant traces based on an analyst's current process. In a follow-up study, Zhong et al. [149] use the same tracing method but replace the RNN with a centroid-similarity matching method, which is faster. However, the centroid calculation algorithm assigns weights to operations based on an analyst's focus, potentially requiring manual tuning for optimal performance. To address this, automatic weight allocation can be achieved through additional offline graph structural analysis, ensuring the retrieval system's run-time efficiency.

6.2 Augmentation-based Alert Fatigue Mitigation Solutions

AI-enabled augmentation can help alleviate the problem of alert fatigue in SOC. Augmentation aims to provide human analysts with supplementary information or context, facilitating their decision-making process. In this study, we categorise augmentation methods into three main categories: (i) Visualisation, (ii) Sonification, and (iii) Explanation. Figure 3 and Table 6, respectively, categorise and summarise augmentation-based alert fatigue mitigation solutions.

6.2.1 Visualisation. Visualisation tools help analysts understand complex data, identify patterns and anomalies, and respond to potential threats more effectively. An efficient SOC team relies on proper visualisations. Reducing the number of dashboards or windows that analysts must switch between can also help reduce alert fatigue. A recent usability study by McRee [82] finds that visual alerts are more effective than other methods in improving detection and response. Nevertheless, delivering concise, centralised information in a unified dashboard remains a challenge.

Various network traffic visualisation tools have been used by security analysts since the early 2000s [127]. Recently, there has been a focus on methods that help analysts quickly understand the context of alerts [17]. While these tools were not specifically designed to reduce alert fatigue, they can enhance analysts' ability to identify anomalies and understand complex data patterns, thereby reducing cognitive load.

Nadeem et al. [92] propose SAGE, an alert-driven attack graph generator that uses temporal and probabilistic dependencies between alerts to provide insights into past and potential attacks. Similarly, Rabb et al. [109] propose an interactive process tree analysis visualisation for exploring

Table 6. A Summary of Augmentation-based Alert Fatigue Mitigation Solutions

Method (Name; Technique)	Dataset	Performance (Metric)	Remarks (Pros; Cons; Future directions)
N: Context-based Alerts [17] T: CoNTA	Real SMB2 Network Traffic Data	Not Specified	P: Supports quick probe into alert collections C: Scalability issue F: Refine classifier integration
N: SAGE [92] T: Constructs attack graphs	CPTC-2018, CPTC-2017, CCDC-2018	Compresses 330K Graphs into 93 Attacks	P: Reduces alert fatigue C: Imperfect IDS F: Enhance model interpretability
N: Process Tree Analysis [109] T: Augmented visualisation of a process tree	Endpoint Sensor	Not Specified	P: Aids in understanding process creation sequence C: Steep learning curve F: Validate in real-life scenarios
N: Graph Feature Embedding [16] T: Feature embeddings	Network Data	Not Specified	P: Enables automatic aggregation of related alerts C: Requires manual investigation F: Train models with additional event fields
N: SHAP and LIME for SOC [39] T: SHAP and LIME	UNSW-Nb15	Focuses on Explainability	P: Provides insights and highlights features C: Requires understanding of ML-based alerts F: Tailor and enrich XAI methods
N: MATE [73] T: Dynamically links and organises alerts	Real-world Alerts	Not Specified	P: Reduces alert volume C: Lacks a graphical interface F: Develop a user interface for MATE
N: Explanation using Symbolic Methods [55] T: Combining symbolic and sub-symbolic methods	Events in Hardware Demonstrator	84% (ROC) 66% (FP Reduction)	P: Reduces FP alerts C: GNNs lack transparency F: Real-time analysis; scalability on larger systems
N: Sonification in SIEM (S-SIEM) [9] T: Experimental setting with/without sonification	CIC-IDS 2017	50% Detection time reduction	P: Enables peripheral monitoring of sound C: Potential for overload/confusion F: Validate utility of sonification systems

the temporal behaviours of processes. Burr et al. [16] argue that visually modelling attacks is effective, especially for APT campaigns, and use feature embeddings from network event logs to build a weighted graph. We posit that incorporating improved categorical feature embeddings can enhance alert correlations and reduce alert numbers. Future research should explore sophisticated embedding techniques, such as **graph neural networks (GNNs)**, to capture complex relationships. Combining graph embedding with DL can help identify attack patterns that cannot be detected using conventional methods.

6.2.2 Explanation. AI-powered systems are increasingly mediating decision-making processes in SOC. To enable informed and responsible decision-making, these systems must provide clear and actionable explanations. This can enable SOC analysts to understand and resolve alerts quickly, thereby reducing the amount of time spent on triage and decreasing alert fatigue. Thus, AI-powered systems must provide clearer and more actionable explanations.

Ehsan et al. [37] argue that explanations in human interactions are socially contextual. While AI systems are embedded in socio-organisational settings, **Explainable AI (XAI)** often focuses on algorithms. To address this, they propose a socially situated XAI framework, **Social Transparency (ST)**, which incorporates socio-organisational context into AI explanations. Developed through interviews with AI users and SOC analysts, ST aims to enhance trust, decision-making, collective actions, and holistic explainability by integrating social and organisational factors. Similarly, Eriksson and Grov [39] conducted a user study on SOC analysts to explore how XAI can explain ML-generated alerts. They argue that the lack of transparency in ML hampers trust and adoption in SOC. Their study focuses on two XAI techniques, SHAP and LIME, showing that XAI can highlight important features and information for alerts, enhancing trust, transparency, and accountability in AI-mediated decision-making.

MATE, proposed by Lin [73], automatically organises and summarises alerts into cases for prioritisation and investigation. Unlike prior methods that require pivoting between alerts to establish connections, MATE generates interpretable threat candidates with a timeline of activities based on the MITRE ATT&CK framework. This provides SOC analysts with context and

explanations consistent with their knowledge, expediting the triage process. Enhancing MATE's user experience and interface design remains a challenge that warrants further research.

The use of knowledge graphs and GNNs in cybersecurity is growing, as they integrate data from multiple domains using human-understandable vocabularies. However, the lack of transparency in GNN decision-making is concerning, especially given the high incidence of false positives requiring expert triage. To address this, Himmelhuber et al. [55] propose an XAI approach combining symbolic and sub-symbolic methods with domain knowledge to enhance trust management. This method generates intuitive explanations for alerts, reducing false positives. However, it tends to produce more false negatives, potentially missing critical alerts. Balancing this tradeoff or finding alternatives is crucial.

6.2.3 Sonification. SOC analysts often work extended hours, including night shifts, monitoring multiple screens for prolonged periods. Timely identification of malicious activity is crucial for organisational security. To enhance analysts' response to alerts, researchers have proposed sonification (non-speech audio) to communicate or contextualise data. Louise et al. [8] conducted a survey and interviews with SOC analysts to explore sonification's applicability. The study found that sonification could benefit analysts, especially when multitasking, working on multiple screens, or away from the computer. They identified settings where sonification could reduce alert fatigue. Subsequently, Louise et al. [9] implemented and evaluated a sonification system in a real-world S-SIEM. Results showed improved performance with sonification, particularly when monitoring a secondary activity while performing a primary task, thus reducing alert fatigue. However, sonification is not foolproof; multitasking participants showed reduced precision in detecting abnormal traffic, suggesting that auditory cues may sometimes mislead analysts. A major challenge is designing sonification techniques that adapt to various mediums and remain usable for extended periods without causing listener fatigue.

6.3 Collaboration-based Alert Fatigue Mitigation Solutions

In the context of SOC, human-AI collaboration aims to leverage the complementary strengths of human analysts and AI systems to detect and respond to security incidents. We categorise these solutions into two categories: (i) Active Learning and (ii) Interactive Conversational Interfaces. In Figure 3 and Table 7, we provide a systematic categorisation and summary of collaboration-based solutions, respectively.

6.3.1 Active Learning. AL is an ML approach that selects the most informative data points to label, aiming for better predictive performance with fewer labelled points. Sopan et al. [132] propose a model that integrates feedback from SOC analysts to improve performance. They used real alerts labelled by analysts to enhance accuracy. The system displays its prediction for each alert with an explanation, helping analysts categorise alerts and understand the model's predictions. While this approach is promising, further improvement could involve more collaborative dialogue between AI and analysts. Chung et al. [23] suggest using **interactive machine learning (iML)** to enhance collaboration between domain experts and algorithms for detecting data exfiltration attacks, reducing false positives and analyst fatigue. They used AL with the LightGBM framework, which is more resource-efficient than neural networks. However, DL methods offer advantages like greater generalisability. With increasing GPU availability and affordability, exploring DL methods is a potential future research direction.

6.3.2 Interactive Conversational Interfaces. **Interactive Conversational Interfaces (ICI)**, such as chatbots, have become effective tools for user-computer communication using spoken language, text, or other modalities. These interfaces simulate human conversation using AI algorithms

Table 7. A Summary of Collaboration-based Alert Fatigue Mitigation Solutions

Method (Name; Technique)	Dataset	Performance (Metric)	Remarks (Pros; Cons; Future directions)
N: Feedback with Explainability [132] T: ML model to classify alerts	Security Alerts	98.50% (ML Success)	P: Reduces analyst workload C: Model may not be sufficient for all alert types F: Incorporate more features, improve the model
N: Interactive ML [23] T: iML framework using AL	CERT Dataset	97.06% (TPR) 0.0057% (FPR)	P: Prunes false alarms C: Performance/compatibility tradeoff F: Make outputs more explainable
N: Intelligent SOC Chatbot [107] T: Creating a chatbot for SOC's	Simulated Setup	Not Specified	P: Automates incident ticket creation C: Needs real-world implementation F: Develop the chatbot to detect incidents
N: Ask Me Anything Chatbot [43] T: Artemis, a conversational interface to EDR event data	Simulated Setup	Not Specified	P: Reduces alert fatigue C: Improves natural language understanding F: Enabling intuitive collaboration among users
N: Insights through Summarisation [34] T: Chatbots for information security practitioners	Simulated Setup	Not Specified	P: Provides quick, accurate answers C: Limited chatbot knowledge F: Improve hand-off between AI and humans

to understand user input and provide appropriate responses. Recently, ICIs have gained attention in cybersecurity. Kowalski et al. [67] found that security specialists have a positive attitude toward using chatbots for security. Perera et al. [107] proposed a chatbot that consolidates data and procedures, streamlining processes and recommending investigation paths, significantly reducing task time. Filar et al. [43] suggested a conversational interface to help inexperienced analysts understand their security environment, reducing the workload of experienced analysts. Dutta et al. [34] proposed a scalable chatbot that reduces cognitive load and stress by gathering recent information on security breaches. Advances in conversational agents such as OpenAI’s ChatGPT, Meta’s Llama, Microsoft’s Co-pilot, and Google’s Gemini make chatbot-assisted SOC analysts a promising prospect. However, these systems face challenges, such as vulnerability to sophisticated cyber attacks [129], data privacy concerns, and the difficulty of capturing non-textual analyst interactions.

7 Industry Initiatives to Mitigate Alert Fatigue

This section explores industry standards and informal, de facto practices aimed at mitigating alert fatigue in SOC's.

STANDARDS AND REGULATIONS. Several standards, frameworks, and guidelines from organisations such as the **National Institute of Standards and Technology (NIST)** and **MITRE** address various aspects of security operations. For instance, NIST SP 800 – 53 emphasises effective incident response and monitoring, while NIST SP 800 – 137 and NIST SP 800 – 92 address continuous monitoring and log management. ISO/IEC 27001 and 27002 provide general guidance on the management of security events and incidents. However, none of them provide specific recommendations for mitigating alert fatigue. Other frameworks, such as the **NIST Cybersecurity Framework (NIST-CSF)** [94] and the UK’s **National Cyber Security Centre (NCSC)** guidelines [93], indirectly contribute to reducing cognitive overload by focusing on building robust and efficient SOC's. MITRE’s 11 strategies for developing world-class SOC's [65] indirectly help mitigate alert fatigue. In sum, while existing standards and frameworks such as NIST-CSF [94], MITRE ATT&CK [84], SOC-CMM [131], and SOC-Class [85] focus on setting up a SOC and enhancing SOC capabilities, they often overlook the human factor. Addressing alert fatigue requires a more nuanced approach, involving optimising alert thresholds, improving alert quality, and implementing human-centred design principles. These aspects are more commonly addressed by industry-specific tools and practices rather than by broad, overarching standards and frameworks.

DE FACTO INDUSTRY PRACTICES. The SOC industry has introduced various tools, platforms, and strategies to tackle common challenges, including alert fatigue. These practices, often highlighted in annual SOC industry surveys [26, 56], have become de facto standards adopted

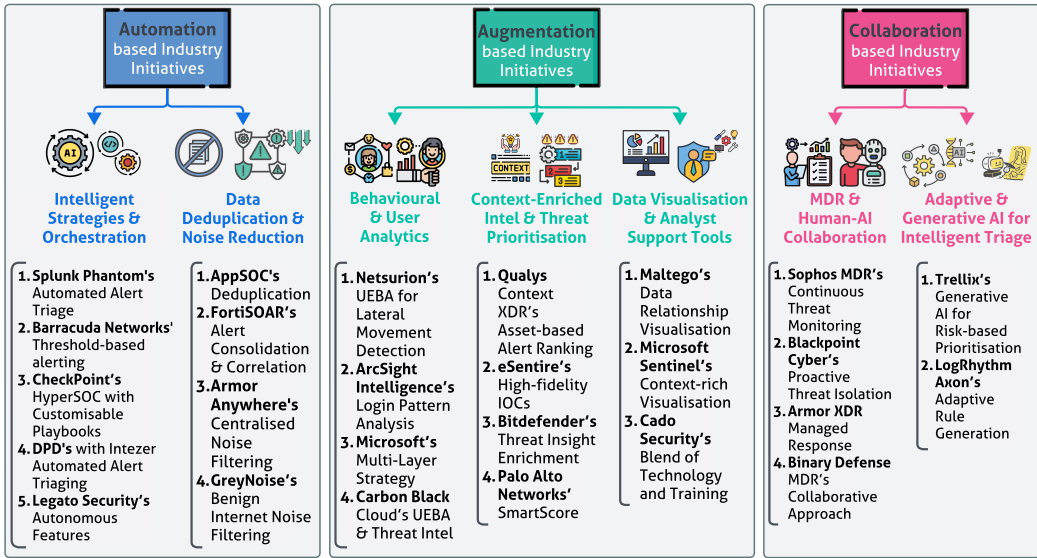


Fig. 4. A categorisation of industry initiatives for alert fatigue mitigation.

by industry players. The following sections examine prominent industry solutions to mitigate alert fatigue in SOCs, as summarised in Figure 4. Each approach is analysed in terms of its methodology, outcomes, limitations, and potential areas for future improvement. To achieve this, we conducted a systematic search of industry-developed SOC tools, employing a methodology akin to that used in our literature review. First, we defined inclusion criteria to ensure relevance and focus: The tools had to be SOC-related and specifically aimed at minimising alert fatigue in SOCs. Using Google Search as the primary tool, we combined keywords such as “alert fatigue” and “alarm fatigue” with “SOC(s),” “security operation(s) centre(s),” or “security operation(s) center(s)” to formulate structured search queries. For example, we employed combinations such as Query 1: “alert fatigue” “SOC” and Query 2: “alarm fatigue” “SOCs” to identify potential tools. The search results were then systematically screened, and SOC tools meeting the predefined inclusion criteria were selected for further analysis. The selected tools were evaluated through the A^2C lens. The findings are presented in the following subsections.

7.1 Automation-based Initiatives

Automation solutions streamline SOC workflows by reducing manual tasks, consolidating data, and filtering unnecessary alerts. By automating repetitive tasks, these solutions enable analysts to concentrate on more critical issues.

7.1.1 Intelligent Strategies and Orchestration. Various tools and strategies exemplify advancements in this domain. For instance, Splunk Phantom, a SOAR tool, streamlines alert triage, enhancing SOC efficiency by minimising human involvement in routine threat management [33]. Similarly, Barracuda Networks employs threshold-based alerting, thereby reducing unnecessary distractions [98]. Check Point's HyperSOC Platform integrates Torq's playbooks to automate repetitive alert responses, thereby enhancing operational efficiency. AI-driven contextual understanding could further improve the handling of complex scenarios [137].

Microsoft employs ML, automation, and UEBA to reduce false positives and enhance analyst productivity. While effective, frequent model adjustments are required to maintain effectiveness.

Self-adaptive ML systems could make this approach more scalable and sustainable [120]. Similarly, Intezer's AI-driven triage system automates the initial alert assessment, filtering out low-priority cases. However, its performance hinges on high-quality input data, and its utility could be expanded with broader datasets and continuous learning mechanisms [80]. Legato Security's Autonomous SOC Platform offers a similar approach by autonomously managing routine security tasks, though additional refinements in contextual threat prioritisation could further reduce manual involvement [60].

DPD's integration of Intezer solutions focuses on automating alert triaging to prioritise high-stakes incidents, with adaptive learning mechanisms offering potential improvements in detection accuracy and minimising misclassifications [61]. Fortra Tripwire employs a risk-scoring system to prioritise alerts, allowing analysts to concentrate on higher-risk cases while filtering out lower-priority notifications [138]. Although these automation solutions alleviate manual workloads, they often depend on predefined rules. Future advancements should focus on adaptive algorithms capable of dynamically adjusting to real-time threat intelligence, enhancing flexibility and responsiveness.

7.1.2 Data Deduplication and Noise Reduction. Data deduplication and noise reduction techniques are critical in minimising alert fatigue by consolidating duplicate alerts and filtering out non-actionable notifications. Tools like AppSOC effectively reduce redundancy by deduplicating alerts across multiple sources, streamlining the alert management process [14]. FortiSOAR takes this a step further by consolidating and correlating alerts to prioritise significant events, helping security teams focus on critical threats [46]. Similarly, Armor Anywhere centralises noise filtering and alert handling across different environments, reducing distractions for analysts [130]. GreyNoise, integrated with Panther Labs, suppresses irrelevant internet "noise," allowing SOC teams to concentrate on actionable alerts. However, this approach requires constant updates to address changing patterns of benign internet activity. Enhancements such as real-time adaptive filtering could further improve its ability to respond to evolving activity types [49].

While these techniques are highly effective in reducing workload and improving alert prioritisation, they may inadvertently overlook unique or rare threats if the deduplication criteria are too rigid. To address this limitation, future advancements should focus on adaptive algorithms capable of dynamically adjusting deduplication criteria. Such approaches would ensure that meaningful but uncommon events are not filtered out, striking a balance between reducing noise and maintaining sensitivity to rare yet critical threats.

7.2 Augmentation-based Initiatives

Augmentation solutions enhance SOC analysts' capabilities by providing context, prioritising threats based on risk, and leveraging advanced analytics. These tools help improve both efficiency and decision-making quality.

7.2.1 Behavioural and User Analytics. Behavioural and UEBA solutions are designed to identify anomalies by establishing behavioural baselines and leveraging ML to reduce false positives. For example, Netsurion's UEBA capabilities are adept at detecting stealthy attacks by identifying lateral movement within networks [97]. Similarly, ArcSight Intelligence focuses on identifying insider threats by analysing anomalous login patterns, providing critical insights into potentially malicious activities [45]. Microsoft's multi-layered strategy integrates ML with UEBA to dynamically categorise alerts, significantly reducing false positives. However, this approach requires frequent model adjustments to remain effective, and adopting self-adaptive learning models could enhance its capacity to continuously update baselines, ensuring accuracy in dynamic environments [120].

Carbon Black Cloud integrates UEBA with threat intelligence to detect behavioural deviations, adding another layer of detection capabilities. Despite its effectiveness, implementing such solutions often requires substantial computational resources, limiting their accessibility to smaller SOC [25]. Future work could focus on resource-efficient behavioural models to improve usability across organisations of varying sizes. Also, incorporating adaptive and continuous learning models could further refine their effectiveness, ensuring consistent performance even as environments evolve.

7.2.2 Context-enriched Intelligence and Threat Prioritisation. Context-enriched intelligence solutions enhance threat prioritisation by incorporating contextual data and evaluating asset criticality. Tools such as Qualys Context XDR rank alerts based on the importance of the associated assets, ensuring that critical threats receive timely attention [19]. Similarly, eSentire provides high-fidelity **Indicators of Compromise (IOCs)** to enable rapid and precise responses to emerging threats [40]. Bitdefender enriches threat intelligence with actionable insights, supporting better decision-making during incident response [59]. Palo Alto Networks' SmartScore dynamically scores incidents by leveraging contextual data, optimising alert triage and enabling SOC teams to focus on the most pressing issues [121].

While these solutions are effective at prioritising threats, their performance is heavily reliant on the quality and accuracy of the contextual data they process. This reliance can limit their reliability in complex or rapidly changing environments where contextual information may be incomplete or outdated. Integrating adaptive ML techniques could address these limitations by dynamically refining prioritisation criteria and enabling more nuanced alert management. Such advancements would allow these tools to remain effective even in evolving threat landscapes.

7.2.3 Data Visualisation and Analyst Support Tools. Visualisation and support tools play a vital role in assisting SOC teams by presenting clear and structured views of data patterns and relationships. These tools enhance situational awareness and reduce cognitive load, enabling analysts to make informed decisions more efficiently. For instance, Maltego's visualisation tools allow analysts to quickly interpret connections among data points, improving response times. However, their steep learning curve may hinder adoption, highlighting the need for more user-friendly interfaces and tailored training resources to enhance usability [111]. Similarly, Microsoft Sentinel provides context-rich visual insights that facilitate rapid decision-making in prioritising and addressing threats effectively [69]. Cado Security combines advanced technological tools with continuous training programs to empower analysts and alleviate alert fatigue [119]. While these tools improve threat comprehension and decision-making, their effectiveness often relies on dedicated training for analysts due to a lack of usability considerations. Research should prioritise usability by developing more intuitive designs and comprehensive training programmes, enhancing accessibility and adoption.

7.3 Human-AI Collaboration-based Initiatives

Human-AI collaboration solutions combine automation with human oversight, enabling expert-driven threat hunting and response while reducing alert fatigue. These solutions empower SOC analysts by offloading routine tasks to automated systems while leveraging human expertise for more complex issues.

7.3.1 Managed Detection and Response (MDR) and Human-AI Collaboration. **Managed Detection and Response (MDR)** solutions combine automation with expert-driven insights to enhance SOC operations. For example, Sophos MDR provides continuous threat monitoring and automated alert triage to streamline incident management with expert oversight [133]. Blackpoint Cyber's

MDR solution extends these capabilities by proactively isolating threats while handling alert triage with experts, offering robust threat containment [27]. Similarly, Armor **Extended Detection and Response (XDR)** integrates managed response with continuous workflow optimisation from expert insights to improve SOC efficiency [6]. Binary Defense MDR takes a collaborative approach, working alongside in-house teams to continuously optimise detection and response strategies [30]. While MDR solutions address critical challenges such as staffing shortages, they often depend on human expertise to handle high-level incidents, which can pose scalability issues. Future research could focus on innovative training methods and advanced tools to enhance scalability and cost-efficiency, ensuring that MDR solutions remain viable across organisations of varying sizes and resources.

7.3.2 Adaptive and Generative AI for Intelligent Triage. Adaptive and generative AI solutions play a critical role in dynamically adapting to evolving threat patterns and user feedback, enabling efficient prioritisation and triage of alerts based on contextual insights and assessed risk. For example, Trellix's generative AI solution based on LLMs directs analysts' attention to high-impact threats by prioritising incidents according to their associated risk levels [15]. LogRhythm Axon enhances triage accuracy by employing adaptive rule generation that adjusts based on contextual information and experts' needs [24]. These AI-driven systems significantly reduce manual workloads and enhance alert prioritisation. However, their reliance on high-quality input data and frequent model retraining presents challenges in maintaining consistent performance. Integrating real-time feedback mechanisms could enable continuous adaptation to new threats and user inputs, further improving their effectiveness in dynamic cybersecurity environments.

7.4 Real-world Examples and Case Studies

This section presents five case studies on mitigating alert fatigue and enhancing security operations. These case studies illustrate how automation and contextual augmentation enhance SOC performance.

- (1) **OTORIO and a U.S. Oil Refinery [105]:** OTORIO assisted a U.S.-based oil refinery in mitigating alert fatigue by incorporating operational technology risk context into its security framework. This augmentation significantly reduced false positives, distinguishing critical threats from non-critical ones and enabling security teams to prioritise alerts effectively, addressing the *High False-alarm Rate* challenge. By contextualising threats, the approach streamlined threat assessment, ensuring analysts focused on high-risk issues rather than being overwhelmed by excessive, uncontextualised alerts, thereby addressing *Disconnected and Overloaded Dashboards*.
- (2) **Legato Security and Intezer's Autonomous SOC [60]:** Legato Security enhanced SOC operations by deploying Intezer's Autonomous SOC platform, which automated key processes to improve scalability, thereby addressing *Inefficient SOPs*. The system filtered out noise, prioritised genuine threats, and optimised threat analysis and response workflows. Consequently, operational efficiency increased, incident resolution times improved, and analysts experienced a notable reduction in false positive fatigue, tackling the *High False-alarm Rate* issue.
- (3) **OTAVA and a Technology Solutions Company [104]:** OTAVA implemented a customised SIEM and SOC solution for a technology firm, reducing monthly alerts from over 700 million to just 18 actionable events. This drastic reduction underscores the burden of false positives on analysts, addressing the *High False-alarm Rate*. By centralising and optimising alert management, the solution ensured only critical threats reached security personnel, mitigating

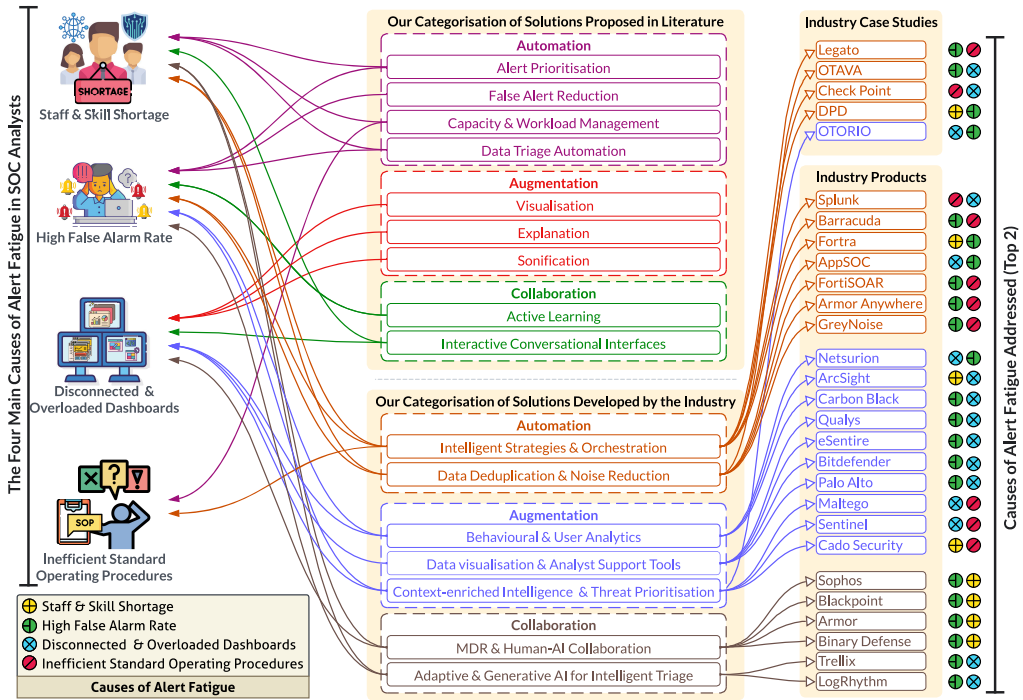


Fig. 5. Mapping of proposed solutions from the literature (top) and industry practices (bottom) to the corresponding causes of alert fatigue in SOCs. Solutions are categorised through the A²C lens. On the right, industry case studies and products are individually mapped to the specific alert fatigue causes they address.

inefficiencies caused by fragmented security tools and uncorrelated alerts (*Disconnected and Overloaded Dashboards*).

- (4) **Check Point and Torq's HyperSOC [137]:** Check Point integrated Torq's HyperSOC platform to automate alert investigation, triage, and remediation, significantly reducing alert fatigue and alleviating *Disconnected and Overloaded Dashboards* issue. Automated workflows minimised manual handling of false alarms, streamlining security operations and addressing *Inefficient SOPs*. By eliminating redundant processes, the system enhanced operational efficiency, allowing analysts to focus on genuine security alerts and reducing the *High False-alarm Rate*.
- (5) **DPD Poland and Autonomous SOC Automation [61]:** DPD Poland adopted Intezer's Autonomous SOC platform to automate endpoint security alert triage, investigation, and remediation, saving over 2,500 hours annually. This automation addressed *Inefficient SOPs* by reducing manual alert reviews and enabling analysts to focus on real security threats, thereby mitigating the *High False-alarm Rate*. Additionally, the automation of Tier 1 SOC tasks alleviated skill shortages and optimised resource allocation, allowing security teams to concentrate on strategic security initiatives and addressing the *Staff and Skills Shortage*.

8 Mapping Alert Fatigue Causes to Solutions from Literature and Industry

This section examines the relationship between the causes and mitigation strategies for alert fatigue, as illustrated in Figure 5. By analysing academic research and industry practices, we map causes to solutions, identify overlaps and divergences, and highlight research gaps. Our analysis

shows that industry solutions are more sophisticated and polished, underscoring the industry's role in driving innovation in this domain.

AUTOMATION SOLUTIONS: ADDRESSING OPERATIONAL OVERLOAD. Automation is the most prominent solution in both academic literature and industry initiatives for mitigating alert fatigue. Automation streamlines routine tasks such as alert triage, prioritisation, and data deduplication, addressing three primary causes of alert fatigue: staff and skill shortages, high false-alarm rates, and inefficient SOPs. In the literature, approaches such as alert prioritisation frameworks (e.g., MADE [103]) and false alert reduction techniques (e.g., NoDoze [54]) focus on reducing the cognitive load on analysts. Similarly, industry tools such as Splunk Phantom and Microsoft Sentinel leverage SOAR capabilities and ML-driven triage to automate repetitive workflows, enhancing SOC efficiency [33, 120]. A notable strength of these industry solutions is their practicality and scalability.

A central challenge in automation-based alert fatigue mitigation is balancing recall (minimising false negatives) and precision (minimising false positives). Preventative security tasks, such as intrusion detection, malware detection, and phishing prevention, by their nature, prioritise recall to ensure rare but critical threats are detected [64, 95, 147]. However, this frequently results in excessive false positives and analyst overload [3, 10]. Conversely, operational security tasks—such as SOC alert triage, authentication monitoring, and fraud detection—prioritise precision to minimise irrelevant alerts [56, 86]. Excessive false positives contribute to cognitive overload, increasing the risk of analysts overlooking real threats [35, 83]. Methods prioritising recall, such as anomaly detection-based intrusion detection [147] and threat intelligence integration [140], aggressively flag threats to prevent missed incidents, but this can generate excessive false positives and contribute to alert fatigue [95]. Conversely, precision-focused strategies, including risk-based alert prioritisation [103, 123] and supervised machine learning-based alert suppression [10, 96], filter low-confidence alerts to improve efficiency. However, this can inadvertently suppress alerts indicative of low-frequency, high-impact attacks, potentially allowing critical threats to go unnoticed [53, 54]. Future research should explore hybrid models that integrate detection with human-in-the-loop decision-making to dynamically balance this tradeoff.

Academic literature and industry approaches differ in adaptability. While literature-based methods emphasise innovative techniques such as RL, these approaches often rely on curated datasets and lack real-world validation. In contrast, industry solutions prioritise scalability and user-friendliness but often depend on predefined rules and thresholds, limiting adaptability to novel or evolving threats. Both academic research and industry practice reveal a critical need for adaptive SOC solutions that dynamically adjust mitigation strategies, including precision-recall thresholds, based on contextual risk assessments. Additionally, robust benchmarking methodologies that assess automation strategies in real-world SOC environments could bridge the gap between academic advancements and industry deployment.

AUGMENTATION SOLUTIONS: ENHANCING ANALYST CAPABILITIES. Augmentation-based solutions enhance SOC analysts' ability to process data, interpret alerts, and prioritise threats. These solutions directly address challenges posed by disconnected and overloaded dashboards while indirectly mitigating staff and skill shortages by enhancing personnel productivity. In the literature, visualisation tools such as SAGE [92] and graph feature embeddings [16] improve situational awareness by grouping related alerts and highlighting high-risk events. Similarly, sonification systems like S-SIEM [9] provide auditory cues to help analysts monitor threats while multitasking. XAI techniques (e.g., SHAP and LIME [37]) provide interpretability to AI-driven threat assessments, assisting analysts in validating system decisions.

Industry solutions such as Microsoft Sentinel and Maltego adopt similar approaches by offering interactive data visualisation and context-rich insights [69, 111]. Industry tools also incorporate

asset criticality scoring, enabling analysts to focus on high-priority threats. While both literature and industry highlight the value of reducing cognitive load, their focus areas diverge. Academic research tends to emphasise novel visualisation and AI explainability methods, while industry solutions integrate these insights into operational workflows with a focus on usability and threat intelligence fusion. A key challenge in augmentation solutions is the steep learning curve of advanced visualisation and interactive AI tools. Future research should focus on designing adaptive interfaces that provide personalised recommendations based on analyst expertise. Developing real-time collaborative dashboards could enhance team-based decision-making and reduce overreliance on automated recommendations.

HUMAN-AI COLLABORATION SOLUTIONS: LEVERAGING COMPLEMENTARY STRENGTHS.

Human-AI collaboration mitigates alert fatigue by integrating automation with human expertise. This category addresses all four primary causes of alert fatigue, albeit with varying emphasis. Collaboration-based solutions effectively address staff and skill shortages and high false-alarm rates, where human oversight is crucial for refining AI-driven processes. In the literature, approaches such as Human-in-the-loop ML frameworks [23] allow analysts to provide real-time feedback to improve AI-driven alerts, and interactive SOC chatbots [107] facilitate human-AI dialogue to contextualise alerts and accelerate triage. Industry initiatives, such as Sophos MDR and Trellox's generative AI systems, similarly emphasise the integration of automation with expert-guided threat hunting [15, 133]. These solutions reduce manual workloads while enabling expert-driven threat hunting and incident response.

Both literature and industry recognise the importance of human feedback in refining AI-driven SOC solutions. However, literature-driven approaches emphasise AI explainability and AL, whereas industry tools prioritise real-time efficiency and workflow integration. Tailoring XAI techniques to SOC workflows could improve trust and usability, ensuring that rare but critical threats receive adequate attention while reducing unnecessary workload. Additionally, a key gap in both domains is the lack of robust frameworks for measuring human-AI collaboration effectiveness. Future research should develop metrics to evaluate: (i) shared situational awareness (e.g., how well AI recommendations align with analyst decisions), (ii) collaborative decision accuracy (e.g., how often AI-assisted alerts lead to correct SOC responses), and (iii) adaptive precision-recall balancing (e.g., how AI models dynamically adjust based on analyst input).

9 Discussion and Future Research Directions

This section outlines **research opportunities (RO)**, **key challenges (KC)** in realising those ROs, and **actionable recommendations (AR)** for mitigating alert fatigue in SOCs. As shown in Figure 6, we propose 12 research directions leveraging AI-driven automation, augmentation, and human-AI collaboration, including adaptive AI, explainable systems, and gamified training. We identify 17 key challenges, including algorithmic bias, cognitive load, system integration, and privacy compliance. The research opportunities and challenges exhibit a many-to-many relationship. We also recommend cross-disciplinary collaborations across 10 fields, including ML, AI, **human-computer interaction (HCI)**, **cognitive science (CoS)**, and **behavioural science (BeS)** to guide stakeholders, including researchers and practitioners. Finally, we propose six actionable recommendations derived from the identified research opportunities and challenges.

9.1 Research Opportunities

RO1: COLLABORATIVE CAPACITY MANAGEMENT. Advanced ML techniques, such as AL and RL with human feedback, support capacity management in SOCs by learning from past workloads, analyst expertise, and fatigue levels, effectively balancing tasks and preventing individual overburdening. Effective AI decision-making in SOCs necessitates balancing system autonomy with human



Fig. 6. Future research directions focusing on the four causes of alert fatigue: Explores opportunities, identifies key research challenges in realising these opportunities, and highlights the relevant research communities equipped to tackle these challenges.

oversight, especially in high-risk scenarios [128]. Research should focus on criteria for when AI systems should defer to human judgement, enabling analysts to make informed decisions in complex scenarios [87, 88]. Future research should explore optimisation algorithms for fair task distribution, integrating real-time metrics like cognitive load for better workload management.

RO2: AI-ASSISTED TRIAGE. Automated triage aims to reduce alert volumes, but current methods tend to over-prioritise false positives, requiring human oversight for complex cases [3, 150]. AI systems capable of automating routine triage while deferring complex cases to human analysts represent a promising area of development. By learning from analyst responses, these systems can reduce manual effort over time. Future research should enhance AI's capacity to generalise from limited human feedback and manage novel alert scenarios while minimising biases. Conversational AI could further streamline the triage process with natural language interfaces.

RO3: GAMIFICATION FOR TRAINING. Gamification has emerged as an effective training tool, motivating engagement and retention through interactive methods [118]. AI-powered environments like CyberGYM [71] can adapt training based on each analyst's progress, fostering a deep understanding of AI workflows in SOC contexts. Such training can bridge the skill gap and support a smoother transition to AI-powered SOC operations. Future research should explore integrating real-time cognitive load monitoring to ensure training environments remain engaging and effectively mitigate fatigue during skill development. Additionally, developing performance metrics is crucial to assess the long-term impact of gamified training on SOC performance, skill retention, and cognitive engagement.

RO4: ADAPTIVE AI. Continuous learning and lifelong learning techniques can enhance AI systems' ability to adapt to new threat types without frequent retraining [29]. These models leverage feedback from analysts to refine alert classification over time, reducing false positives

and improving SOC efficiency. Future research should focus on creating lightweight and scalable adaptive AI systems that can function effectively in dynamic threat environments.

RO5: CONTEXT-AWARE ALERT PRIORITISATION. Alert prioritisation based on contextual information, such as historical threat patterns, analyst preferences, and organisational risk profiles, represents a promising direction. Hybrid approaches combining ML with domain-specific knowledge can improve accuracy and relevance in diverse SOC settings.

RO6: BEHAVIOURAL SCIENCE-INFORMED MODELS. Integrating insights from BeS into alert prioritisation systems can help manage cognitive load more effectively, as suggested by innovations from Netsurion [97], ArcSight [45], and VMware Carbon Black [25]. Research should investigate real-time cognitive load indicators, such as eye-tracking or physiological monitoring, to optimise alert presentation and minimise analyst strain.

RO7: PERSONALISED ALERT PRESENTATION. AI-driven systems can dynamically tailor alert displays to individual analyst preferences and interaction patterns, reducing cognitive overload. Existing solutions such as Maltego [111] and Microsoft Sentinel [120] demonstrate progress in this direction. By grouping related alerts, filtering out redundancies, and prioritising alerts based on expertise, they can enhance focus and efficiency. Future research could explore the use of ML models that adapt presentation styles based on real-time workload and alert complexity.

RO8: VIRTUAL ASSISTANTS AND LLMs. Advanced NLP tools, including LLMs, can power virtual assistants or chatbots to provide contextual insights and streamline alert triage [65, 113]. These tools enable analysts to interact with alerts using natural language queries, consolidating information from multiple dashboards into a unified interface. Research should focus on tailoring LLMs for SOC-specific tasks, such as summarisation, threat correlation, and decision support.

RO9: INNOVATIVE INTERACTION. Eye-tracking, gesture-based controls, sonification, and **augmented reality (AR)** interfaces offer novel ways to manage alerts intuitively, reducing the strain of navigating cluttered dashboards. For example, sonification was explored by Axon et al. [8], providing an alternative modality for alert interaction. Frameworks such as the **general human-aware interaction (GHAI)** model [148] provide a foundation for integrating AI-driven personalisation with analyst feedback. These systems can optimise interactions by learning analysts' preferences and dynamically adapting alert presentations [70]. Future research could investigate how these paradigms enhance situational awareness and speed up decision-making in high-stress SOC environments.

RO10: PLAYBOOK AND RUNBOOK AUTOMATION. AI can analyse system logs and analyst interactions to generate or update playbooks,³ streamlining response workflows. Tools such as Extractor [114] and AttackKG [72] enable automatic extraction of threat patterns, allowing AI to populate playbooks with current, actionable intelligence. Automated playbooks streamline response workflows, enabling analysts to rely on AI-assisted guidance, which reduces repetitive tasks and operational strain. Future research should develop adaptive systems that update playbooks in real time to stay relevant to evolving threats and operational needs.

RO11: AI-ASSISTED OPTIMISATION. Digital twins and AI-driven simulations can streamline workflows by simulating scenarios, predicting outcomes, and recommending process improvements [32]. Regular updates to playbooks based on current threat intelligence, as demonstrated by solutions like Armor XDR [6], can significantly reduce human workload and improve SOC efficiency. Hybrid intelligence frameworks [148] can enable adaptive SOPs that dynamically blend automation with human decision-making. For instance, these frameworks can ensure that highly

³A runbook details steps for routine SOC activities; a playbook provides a broader strategy guide. For simplicity, we use "playbook" to encompass both.

complex or context-sensitive alerts are deferred to analysts while routine alerts are handled automatically. This adaptive balance ensures that workflows remain robust in dynamic threat environments. Future work could explore integrating real-time threat intelligence and dynamic environmental data to enhance simulation accuracy and utility.

RO12: FRAMEWORKS FOR DYNAMIC COMPLIANCE HANDLING. Developing AI systems that dynamically prioritise alerts based on regulatory requirements and risk contexts could streamline compliance while reducing operational strain. For example, integrating automated compliance checks into playbook updates ensures that SOC's remain aligned with frameworks such as GDPR [41] and HIPAA [139] without excessive manual intervention.

RO13: AI-DRIVEN WORKFLOW OPTIMISATION. The integration of AI into SOC workflows must be strategically designed to maximise its benefits across automation, augmentation, and collaboration. Instead of retrofitting AI into existing processes, novel AI-driven workflows should be developed [62] to streamline SOC operations and improve decision-making. Future research should focus on designing adaptive workflows that ensure AI tools are seamlessly embedded into analyst routines while preserving human oversight where needed [135].

9.2 Key Challenges

KC1: REAL-TIME COGNITIVE LOAD INDICATORS. One of the key challenges is identifying and incorporating cognitive load metrics into AI systems for task allocation and triage. Technologies such as eye-tracking and physiological monitoring, commonly studied in HCI and CoS, must be refined for non-intrusive deployment in SOC's. These indicators would not only help optimise workload but also provide valuable insights into training and capacity management systems.

KC2: ENSURING DECISION TRANSPARENCY. Collaborative AI systems must maintain transparency to ensure analysts' trust [18]. Frameworks like socially situated XAI, developed in ML and AI communities, offer promising solutions to enhance decision transparency [37, 142]. Lack of explainability could lead to resistance in adopting these systems, particularly when analysts are already under stress due to staffing shortages.

KC3: ADDRESSING BIAS IN TASK ALLOCATION. Methods used for collaborative task management are prone to biases, likely leading to uneven task distribution or overlooking less-experienced analysts. This challenge spans ML, BeS, and Psychology, necessitating collaborative efforts to develop bias-mitigation algorithms without compromising efficiency.

KC4: DESIGNING ADAPTIVE TRAINING ENVIRONMENTS. Developing training systems that cater to diverse skill levels and learning paces is a challenge. Gamification must balance engagement with SOC workflow needs and analysts' evolving cognitive states during high-stress training. Behavioural Economics insights are crucial for understanding how motivational systems impact retention and performance.

KC5: DEVELOPING ROBUST SELF-LEARNING MODELS. Self-learning models must balance adaptability and stability to avoid overfitting to specific threat patterns or contexts. They should also be designed to handle novel or unexpected attack vectors without compromising accuracy. Contributions from ML researchers are critical to designing systems that can continuously adapt to changing threat landscapes while remaining robust.

KC6: MINIMISING COGNITIVE OVERLOAD. While improved alert accuracy reduces false positives, analysts may still face cognitive overload due to high volumes of genuine alerts. HCI and CoS researchers can design intuitive interfaces and strategies to optimise alert presentation, enabling analysts to concentrate on the most critical alerts.

KC7: MAINTAINING TRUST IN ADAPTIVE AI SYSTEMS. Analysts may resist relying on AI systems for alert prioritisation unless these systems are explainable and transparent. XAI, a focus area within

ML and AI, offers solutions for delivering clear, interpretable reasoning behind prioritisation decisions [18]. Another avenue is building calibrated trust [81], requiring BeS and HCI researchers to design systems that align with analysts' expectations and cognitive needs.

KC8: METRICS FOR EVALUATING PERFORMANCE. Traditional metrics such as accuracy and F1-score fail to capture the collaborative dynamics and human factors in SOC environments. Researchers from BeS, ML, and HCI must work together to develop holistic metrics that consider task distribution, cognitive load, and overall analyst efficiency.

KC9: ENSURING USABILITY AND INTUITIVE DESIGN. Personalised dashboards and advanced interaction paradigms must prioritise usability to prevent additional cognitive strain. Researchers in HCI and CoS are essential for designing interfaces that balance functionality and simplicity, presenting actionable information while minimising distractions.

KC10: INTEGRATING DISPARATE SYSTEMS. SOC typically rely on a combination of legacy and modern tools. Harmonising these systems into a seamless workflow remains a challenge, as it requires harmonising disparate data formats and ensuring compatibility across platforms. Collaboration between Systems Engineering and Cybersecurity researchers can enable more robust integration strategies.

KC11: DATA PRIVACY AND SECURITY CONCERNS. Personalised alert systems and virtual assistants rely on analyst behaviour and operations data, raising concerns about privacy and security. Research in Privacy and Data Security disciplines is needed to ensure secure handling of sensitive data while maintaining operational effectiveness.

KC12: ADOPTION OF NOVEL INTERACTION PARADIGMS. While technologies such as eye-tracking and AR hold promise, their adoption may face resistance due to cost, usability concerns, or unfamiliarity among analysts. Interdisciplinary research involving HCI, CoS, and BeS can help design systems that align with analysts' expectations and cognitive demands, making them more acceptable and effective.

KC13: BALANCING AUTOMATION AND HUMAN OVERSIGHT. While automation reduces manual effort, over-reliance on automated SOPs can result in missed nuances or context-specific considerations. Research in ML and HCI is essential for designing flexible systems that incorporate human judgement, especially in complex or high-stakes scenarios.

KC14: INTEGRATING TOOLS ACROSS DISPARATE SYSTEMS. SOC typically operate with a mix of legacy and modern systems. Harmonising these tools into a cohesive workflow is a significant challenge, requiring scalable solutions from Systems Engineering to preserve operational continuity during transitions. Collaborative frameworks that combine insights from Systems Engineering and Cybersecurity can enable better integration strategies.

KC15: ADDRESSING REGULATORY COMPLIANCE CONSTRAINTS. Regulations such as GDPR and HIPAA impose strict requirements on data handling and monitoring, often limiting the extent to which SOC workflows can be automated. Legal and regulatory studies research is critical for designing adaptable compliance frameworks for AI-driven systems.

KC16: ADAPTING TO EVOLVING THREAT LANDSCAPES. As threats evolve, SOPs must also adapt dynamically. Ensuring that AI-powered systems can incorporate new intelligence and respond to novel threats promptly requires advanced ML models and continuous updates. Contributions from the Threat Intelligence and Cybersecurity communities are essential for developing systems that maintain relevance in volatile environments.

KC17: BIAS IN AI SYSTEMS. AI-automated SOPs can inadvertently embed biases from historical datasets, reinforcing inefficiencies or inaccuracies in threat detection and response. Collaboration between ML and BeS researchers is necessary to identify and mitigate these biases, ensuring SOPs remain fair and effective.

9.3 Actionable Recommendations

AR1: FOSTER RESEARCH-INDUSTRY PARTNERSHIPS. Collaborating with SOC solution providers can facilitate real-world testing of adaptive dashboards, human-AI workload sharing, and dynamic prioritisation frameworks. Such partnerships can offer insights into aligning systems with analyst workflows to enhance operational resilience.

AR2: IMPLEMENT HUMAN-AI COLLABORATIVE FILTERING. Optimise cognitive and operational efficiency by developing human-AI collaborative filtering systems, where AI manages routine alerts while escalating complex cases to analysts. XAI models that offer interpretable justifications for alert prioritisation can enhance transparency, foster trust, and strengthen human-AI collaboration, particularly in high-stakes scenarios.

AR3: AUTOMATE REGULATORY COMPLIANCE. AI-driven tools for automating routine compliance checks and regulatory reporting can significantly reduce operational burdens. Seamless integration with SOC workflows will enable analysts to focus on critical alerts while ensuring adherence to regulatory frameworks, such as GDPR and HIPAA.

AR4: DESIGN ADAPTIVE DASHBOARDS FOR COGNITIVE LOAD MANAGEMENT. Adaptive dashboards that dynamically adjust to workload, alert severity, and analyst expertise are essential for managing cognitive load. Dashboards should integrate real-time indicators, such as interaction history and eye-tracking, to personalise alert presentation and sustain analyst focus. Usability studies on innovative interaction paradigms, such as gesture-based controls or augmented reality, can further enhance situational awareness and streamline response times.

AR5: ADVANCE CONTEXTUAL ALERT CORRELATION AND FILTERING. Developing AI models capable of consolidating related alerts and correlating them with historical threat data can provide a holistic view of potential incidents. These models will improve situational awareness by reducing false positives and offering meaningful insights.

AR6: ENHANCE INTERDISCIPLINARY COLLABORATION. Addressing these recommendations necessitates contributions from diverse fields, including ML, HCI, CoS, BeS, and Regulatory Studies. Such collaboration ensures SOC solutions remain technologically advanced and human-centric, promoting operational efficiency while reducing analyst fatigue.

10 Conclusion

This article presents a comprehensive examination of alert fatigue in SOC. Our study involved a systematic review of relevant literature on SOC and alert fatigue. We provided a concise overview of SOC, emphasising their evolution over the years. Additionally, we examined the alert validation process in SOC, detailing how it contributes to alert fatigue. From our literature review, we identified four major causes of alert fatigue and categorised existing academic and industry solutions through three perspectives: automation, augmentation, and human-AI collaboration. We further classified these solutions, examined their limitations, and mapped alert fatigue causes to relevant academic and industry solutions. This mapping provides insights into future research directions to address each cause. We also identified broader opportunities, including advancing human-AI collaboration in SOC, and proposed actionable recommendations to mitigate alert fatigue.

IMPACT ON FUTURE WORKS. We believe our survey offers a comprehensive overview of alert fatigue in SOC, crucial for identifying gaps in existing solutions and guiding future research. By highlighting persistent challenges and the need for cross-disciplinary innovation, we encourage researchers and practitioners to explore new strategies for mitigating alert fatigue through automation, augmentation, and collaboration. Our emphasis on human-AI teaming presents a promising direction for future research. Recent advancements in LLMs create new opportunities to enhance SOC efficiency and effectiveness. By examining these trends and their applications in SOC, we

aim to foster further research. Furthermore, the broader applicability of our survey to various C2 domains highlights its impact beyond SOCs. The insights from our survey can inform alert management strategies in high-volume environments, extending their impact on future research. Although alert fatigue in SOCs is not a new challenge, its relevance and the demand for effective solutions persist. With its comprehensive analysis and forward-looking discussion, our survey provides significant value for both current practice and future research.

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Received 4 May 2023; revised 27 February 2025; accepted 4 March 2025